

$$\mathcal{F} = \{\emptyset, \Omega, \{a\}, \{b, c, d\}\} \quad \Omega = \{a, b, c, d\}$$

$$\Omega_0 = \{a, b, c\} \quad \mathcal{F} \cap \Omega_0 = \{\emptyset, \Omega_0, \{a\}, \{b, c\}\}$$

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Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- ✓ (1) (a) Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define $A_n = \{j : j \geq n\}$, $n \geq 1$, $\mu(A) = |A|$ (counting measure). Show that monotone continuity from above does **not** hold for this $\{A_n\} \subset \mathcal{F}$.
 (b) On the same $(\Omega, \mathcal{F})$ and with the same $\{A_n\} \subset \mathcal{F}$, can you specify a *different* measure μ' such that the monotone continuity from above does hold this $\{A_n\} \subset \mathcal{F}$?
 (c) In general, under what conditions does the monotone continuity from above hold?
 (d) In general, under what conditions does the monotone continuity from *below* hold?
 (4 × 10 = 40 points)

- ✓ (2) Let $\mathcal{F}$ be a σ -algebra of subsets of $\Omega \neq \emptyset$. Take $\Omega_0 \subset \Omega$ such that $\Omega_0 \notin \mathcal{F}$. Define $\mathcal{F}_0 = \{F \cap \Omega_0 : F \in \mathcal{F}\}$. Show that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .
 (15 points)

- ✓ (3) (a) Give an example of a function f that is **not** $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.
 (b) Give an example of a function f that is **not** $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable, but $|f|$ is $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.
 (2 × 10 = 20 points)

- (4) (a) Let $\Omega_1 = \mathbb{R}$ and $\mathcal{F}_1 = \{\emptyset, (-\infty, 5], (5, \infty), \mathbb{R}\}$ and let $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $T : \Omega_1 \rightarrow \Omega_2$ be defined as

$$T(x) = \begin{cases} 5, & \text{if } x \leq 0 \\ 0, & \text{if } x > 0 \end{cases}$$

Is T $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable? Justify your answer.

- (b) Let $\{\epsilon_i\}_{i \geq 1}$ be a sequence of *random variables* (i.e. measurable functions) defined on some probability space $(\Omega, \mathcal{F}, P)$ and $\{x_i\}_{i \geq 1}$ be a sequence of real numbers (with $\sum x_i^2 > 0$). For some $\beta \in \mathbb{R}$, define a new sequence of real valued functions on $(\Omega, \mathcal{F}, P)$:

$$Y_i(\omega) = x_i \beta + \epsilon_i(\omega), \quad \omega \in \Omega, \quad i \geq 1.$$

and using the sequence $\{Y_i\}_{i \geq 1}$, define the following ~~two~~ real valued functions on $(\Omega, \mathcal{F}, P)$ for each $n \geq 1$

$$\hat{\beta}_n(\omega) = \frac{\sum_{i=1}^n x_i Y_i(\omega)}{\sum_{i=1}^n x_i^2}, \quad \omega \in \Omega.$$

Argue that $\{Y_i : i \geq 1\}$, $\{\hat{\beta}_n, n \geq 1\}$ are sequences of real valued *measurable* functions on $(\Omega, \mathcal{F}, P)$ (i.e. random variables).

(2 × 15 = 30 points)